Generator-Model Comparison

Deep Research Agent · choosing the generator (decompose + claim-gen) role

2026-06-10

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1. Question & motivation

The generator drives a large share of the pipeline (query decomposition **and** atomic claim generation), so the model in that role matters. The default was **Mistral Large 3**, chosen early. We tested whether it's actually the right call against two leaner candidates — **Llama 4 Maverick (17B)** and **Haiku 4.5** — on the full generator role.

A reframing first: among models **not reserved for the eval panel** (Opus, GPT-OSS 120B, DeepSeek R1 are the panel), **Mistral Large 3 (675B) is already the largest/highest-capability candidate.** So the real question was not "is there something better than Mistral?" but "does Mistral's size earn its keep, or is a leaner model good enough — or better?"

2. Method

For each candidate generator: decompose → retrieve (Exa, **k=5, maxChars=2k**, fixed) → generate claims, across **8 cases** (the 4 brief test cases + 4 additional diverse prompts: RAG-hallucination, ultra-processed-food health, remote-work

productivity, MoE-vs-dense). Each candidate does its **own** decomposition (the whole generator role is under test).

Every claim is scored by a **3-judge panel — GPT-OSS 120B, Claude Opus 4.8, DeepSeek R1** (all independent of the candidates) for: - **atomicity** (0/1): exactly one checkable assertion? - **grounding** (0–2): does the cited evidence support the claim?

Judging is chunked (≤12 claims/call) for reliability; unparseable batches are flagged `unjudged` and excluded from means rather than silently scored 0.

Harness: `experiments/generator_comparison.py` → `experiments/results/generator_comparison/`. Judge = panel (not a candidate), so no self-preference among the three generators. (Caveat: Opus and Haiku are both Claude — a mild self-preference risk for Haiku's scores only; Haiku lost decisively, so it doesn't affect the decision.)

3. Results — 8 cases, 3-judge panel

Grounding (0–2) — the north star:

model	gpt_oss	opus	deepseek	panel	claims
llama4_maverick (17B)	1.13	1.27	1.31	**1.24**	147
haiku_4_5	0.90	0.96	1.05	0.97	287
mistral_large_3 (675B)	0.89	0.92	1.01	0.94	635

Atomicity (0/1): llama **0.80** > haiku 0.74 > mistral 0.73 (panel).

Ranking agreement: all three judges independently rank grounding `Llama > Haiku > Mistral` — unanimous. On atomicity, GPT-OSS and Opus rank Llama first; DeepSeek is the lone dissenter (ranks Mistral first on atomicity, but still ranks Llama first on grounding).

4. Findings

- **The smallest model wins on the metric that matters most.** Llama 4 Maverick (17B) beats Mistral Large 3 (675B) on grounding by a wide, unanimous margin (1.24 vs 0.94).

- **Why — Mistral over-reaches.** Inspection of its low-grounded claims shows confident, *specific* assertions the cited excerpt doesn't support (exact taxonomies, author-years, precise category lists) — it fills detail from parametric memory and staples a citation on. Llama hugs the evidence: fewer, tighter claims that paraphrase what's actually in the passage.
- **A stricter prompt did NOT fix it.** We added an explicit hard rule to `synthesis.py` ("use ONLY information explicitly stated in the passages… a claim that is true but not supported does not belong here"). Mistral still grounded worst (0.94) and still over-generated massively: **635 claims vs Llama's 147** (4×), even hitting a token-limit truncation. The over-reach is structural, not prompt-tunable.
- **For grounded synthesis, bigger is a liability, not an asset** — rich world knowledge becomes a source of unsupported specifics. The lean model that defers to the evidence is what verifiable grounding rewards.

Honest caveats: even Llama's grounding (1.24/2 ≈ 62%) is only moderate — real headroom remains for the verification loop + retrieval, regardless of generator. This is single-vote per judge (no formal Cohen's κ / significance) over n=8; 3-judge unanimity on grounding is a strong directional result, not a proof.

5. Decision

Generator = Llama 4 Maverick. Mistral moves to the **in-loop verifier** role (its world-knowledge suits *judging* better than faithful generation, and the in-loop verifier is the baseline slated for replacement by the fine-tuned DeBERTa anyway). **Haiku is dropped** (worst on both axes).

Model independence holds after the swap (`config.assert_independence()` passes): generator = Llama (meta) ≠ in-loop verifier = Mistral ≠ panel {GPT-OSS, Opus, DeepSeek} ≠ baseline {Perplexity, OpenAI}. Wired in `.env` / `.env.example`.

6. Pipeline changes shipped from this experiment

- **`.env` / `.env.example`**: `BEDROCK_GENERATOR_MODEL` → Llama 4 Maverick; `BEDROCK_INLOOP_JUDGE_MODEL` → Mistral Large 3.
- **`synthesis.py`**: explicit anti-parametric-knowledge grounding rule in the prompt.
- **`llm/bedrock.py`**: default `max_tokens` 1024 → **8192** (1024 truncated claim-list JSON; surfaced because claim generation hadn't run live until this experiment).

7. Reproducibility

```
python -m experiments.generator_comparison   # 3 generators × 8 cases ×
        3-judge panel
```

Outputs: `experiments/results/generator_comparison/{model}.json` + `summary.json`.